CSE 430

Compiler Design Lab

Lab Work 3

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Setup Process:

Step 1:

- Making directory
- Navigating to the directory
- Installing GCC & flex.

```
priom@DESKTOP-H264SP0:~$ mkdir "CSE_430_Lab_Work_3"
priom@DESKTOP-H264SP0:~$ ls
CSE_430_Lab_Work_3
priom@DESKTOP-H264SP0:~$ cd CSE_430_Lab_Work_3/
priom@DESKTOP-H264SP0:~/CSE_430_Lab_Work_3$ sudo apt install flex gcc build-essential -y
[sudo] password for priom:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

Step 2:

- Checking if installation was successful

```
priom@DESKTOP-H264SP0:~/CSE_430_Lab_Work_3$ flex --version
flex 2.6.4
priom@DESKTOP-H264SP0:~/CSE_430_Lab_Work_3$ gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
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This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

priom@DESKTOP-H264SP0:~/CSE_430_Lab_Work_3$
```

Step 3:

- Creating two files `input.c` & `lexer.l` for token classification
- Now we can type code on this 2 files using VS code

```
priom@DESKTOP-H264SP0:~/CSE_430_Lab_Work_3$ touch input.c
priom@DESKTOP-H264SP0:~/CSE_430_Lab_Work_3$ touch lexer.l
priom@DESKTOP-H264SP0:~/CSE_430_Lab_Work_3$ ls
input.c  lexer.l
priom@DESKTOP-H264SP0:~/CSE_430_Lab_Work_3$
```

Input - 1

input.c

input.c > main()

```c
void main(){
    int a, b, c;
    //comment
    int a = b*c + 10;
    if(a!=2)
        a = 0;
}
```

Output - 1

```
priom@DESKTOP-H264SP0:~/CSE_430_Lab_Work_3$ flex lexer.l
priom@DESKTOP-H264SP0:~/CSE_430_Lab_Work_3$ gcc lex.yy.c -o lexer
priom@DESKTOP-H264SP0:~/CSE_430_Lab_Work_3$ ./lexer
1       void    keyword
1       main    identifire
1       (       prenthesis
1       )       prenthesis
1       {       prenthesis
2       int     keyword
2       a       identifire
2       ,       punctuation
2       b       identifire
2       ,       punctuation
2       c       identifire
2       ;       punctuation
3       //comment       single line comment
4       int     keyword
4       a       identifire
4       =       operator(Arithmatic)
4       b       identifire
4       *       operator(Arithmatic)
4       c       identifire
4       +       operator(Arithmatic)
4       10      constant
4       ;       punctuation
5       if      keyword
5       (       prenthesis
5       a       identifire
5       !=      operator(Logical)
5       2       constant
5       )       prenthesis
6       a       identifire
6       =       operator(Arithmatic)
6       0       constant
6       ;       punctuation
7       }       prenthesis
priom@DESKTOP-H264SP0:~/CSE_430_Lab_Work_3$
```

Input - 2

```c
input.c
input.c > main()
void main(){
    int a, b, c;
    //comment
    int a = b*c + 10;
    /*
    line 1
    line 2
    */
    if(a≠2)
        a = 0;
}
```

Output - 2

```
priom@DESKTOP-H264SP0:~/CSE_430_Lab_Work_3$ flex lexer.l
priom@DESKTOP-H264SP0:~/CSE_430_Lab_Work_3$ gcc lex.yy.c -o lexer
priom@DESKTOP-H264SP0:~/CSE_430_Lab_Work_3$ ./lexer
1       void    keyword
1       main    identifire
1       (       prenthesis
1       )       prenthesis
1       {       prenthesis
2       int     keyword
2       a       identifire
2       ,       punctuation
2       b       identifire
2       ,       punctuation
2       c       identifire
2       ;       punctuation
3       //comment       single line comment
4       int     keyword
4       a       identifire
4       =       operator(Arithmatic)
4       b       identifire
4       *       operator(Arithmatic)
4       c       identifire
4       +       operator(Arithmatic)
4       10      constant
4       ;       punctuation
5       /*
    line 1
    line 2
    */  multi line comment
6       if      keyword
6       (       prenthesis
6       a       identifire
6       !=      operator(Logical)
6       2       constant
6       )       prenthesis
7       a       identifire
7       =       operator(Arithmatic)
7       0       constant
7       ;       punctuation
8       }       prenthesis
priom@DESKTOP-H264SP0:~/CSE_430_Lab_Work_3$
```